

DE 424: Practical 10 Report

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Declare that this work presented is your own work. Mention any sources that contributed to work or report
I declare that this report is my own work. This report was written using LaTeX and thus Gemini was used to assist in certain formatting. This includes help to format code listings, mathematical equations and image display. The mathematical derivations and interpretation of results are my own work.

ECSA graduate attributes

Discuss how the GAs were assessed in the module + which parts of the report this is documented. State HOW and WHERE they were satisfied This section may (and should) be brief, but should discuss all aspects of the GAs being assessed. For example, GA1 partially assesses your ability to research literature as part of problem solving, which means that your report must mention and cite (at least) a few sources - can include prescribed resources. Please consult the module framework for a more detailed description of these two GAs as well as how they are assessed in this module.

GA 1: Problem solving

GA 2: Application of scientific and engineering knowledge

1 Problem Description

This project addresses an object tracking task within a discrete grid world inhabited by a latent entity known as a wumpus. The objective is to determine the trajectory of the wumpus across multiple time steps based on inaccurate measurements. At each time step the wumpus attempts to move to an adjacent cell (up, down, left, or right) with equal probability. If the target cell is within the grid boundaries, the wumpus transitions to that location; otherwise it remains in its current position. The exact location of the wumpus is never directly observed. Instead, every cell in the grid generates a binary detection value at each time step, parameterised by:

- True Positive (p_w): The probability that the cell containing the wumpus correctly generates a detection.
- False Positive (p_c): The probability that a cell not containing the wumpus generates an incorrect detection (clutter measurement).

Given the stochastic grid detection data across multiple time steps, the task is to design, implement and evaluate a Probabilistic Graphical Model (PGM) solution. This involves constructing a model that accurately captures the wumpus's transition and observation logic to infer it's latent trajectory across various datasets.

2 Solution Design

The solution was developed iteratively to accomodate the increasing complexity and varying characteristics of each dataset. The PGM design process followed a structured methodology: defining the random variables, constructing the factors, and building a cluster graph for inference. (note: this section must show model structure and show all factors. State and motivate all design choices)

2.1 Dataset 1

The initial model was designed for a 5×5 grid over a series of $T = 10$ time steps. In this scenario, the known parameters $p_w = 0.95$ and $p_c = 0.05$ are constants.

2.2 Define Random Variables

The PGM modelling process first involves defining the random variables (RV's) of the model. A design choice: instead of defining two random variable - one for x and one for y coordinate - a single RV W is chosen so that the wumpus location can be captured in one RV. As such, each cell is numbered from 0 to 24 starting from top left. Capturing the latent wumpus position W^t in a single RV instead of two separate X^t and Y^t variables, results in a significant simplification of the factor construction. Avoids dealing with the complexities of coordinating dependencies between two separate co-ordinates as well as easily sets the probability of illegal moves such as diagonal steps to zero within a single factor. Furthermore, using a single latent variable reduces the total number of RVs within the cluster graph, leading to more efficient message passing and overall faster convergence. The RV's are defined below:

- W^t : Position of the wumpus as time t , $W^t \in \{0, 1, \dots, 24\}$.
- D_w^t : Detection of wumpus at cell $W^t \in \{0, 1, \dots, 24\}$ at time $t \in \{0, 1, \dots, 9\}$.

2.3 Define Factors

3 Code Implementation

Discussion + code snippets

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Description of how you processed the output of solution, the presentation of results and analysis thereof. Report both qualitative and quantitative results.